COHOMOLOGY JUMP LOCI FOR A FLAT PROPER FAMILY FORM A PERFECT COMPLEX ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 6713-6723, immediately following the statement at 6694-6711.

Editorial note. The displayed fiber identity is the span's own step; the three conclusions are read off the terminal lemma.

Allowed standard prerequisites (editorial).

- Base-change perfectness of the pushforward. Lemma lemma-flat-proper-perfect-direct-image-general of this chapter, cited at proof lines 6715-6718. K = Rf_*F is perfect with formation commuting with arbitrary base change.
- Jump loci for perfect complexes on a scheme. Lemma lemma-jump-loci of this chapter, cited at proof lines 6722-6723. Terminal input: base-change compatibility, upper semi-continuity, constructibility of the level sets for the jump function of a perfect complex.

Source reference. Original lemma 0CTM: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 6442–6442. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0D1Y: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 6667–6667. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let Y be a scheme and let X be an algebraic space over Y such that the structure morphism $f : X \to Y$ is flat, proper, and of finite presentation. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module of finite presentation, flat over Y. For fixed $i \in \mathbf{Z}$ consider the function*

$$\beta_i : |Y| \to \{0, 1, 2, \ldots\}, \quad y \longmapsto \dim_{\kappa(y)} H^i(X_y, \mathcal{F}_y)$$

Then we have

(1) *formation of β_i commutes with arbitrary base change,*
(2) *the functions β_i are upper semi-continuous, and*
(3) *the level sets of β_i are locally constructible in Y.*

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. By cohomology and base change (more precisely by Lemma 0CTM) the object $K = Rf_*\mathcal{F}$ is a perfect object of the derived category of Y whose formation commutes with arbitrary base change. In particular we have

$$H^i(X_y, \mathcal{F}_y) = H^i(K \otimes^{\mathbf{L}}_{\mathcal{O}_Y} \kappa(y))$$

Thus the lemma follows from Lemma 0D1Y. □